COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010 / Data Structures and Algorithm	**SECTION:**	CPE21S4
DATE PERFORMED:	July 17, 2026	**DATE SUBMITTED:**	July 17, 2026
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HANDS-ON ACTIVITY NO. 2.1 ARRAYS, POINTERS AND DYNAMIC MEMORY ALLOCATION

1. PROCEDURE OUTPUT

What procedures did you perform?

Table 2-1. Initial Driver Program

SCREENSHOT

HOA2.1_Task2.1_De Vera.cpp > Student > operator=(const Student &)

```
 4   class Student{
23       Student (const Student &copyStudent){
25           studentName = copyStudent.studentName;
26           studentAge = copyStudent.studentAge;
27       }
28
29       // Display Attributes
30       void printDetails(){
31           std::cout << this -> studentName << " " << this -> studentAge << std::endl;
32       }
33
34       // Copy Assignment Operator
35       Student& operator=(const Student &copyStudent) {
36         std::cout << "Copy Assignment Operator was called." << std::endl;
37
38         if (this != &copyStudent) {
39           studentName = copyStudent.studentName;
40           studentAge = copyStudent.studentAge;
41         }
42       return *this;
43       }
44   };
45
46   int main() {
47       Student student1("Roman", 28);
48       Student student2(student1);
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

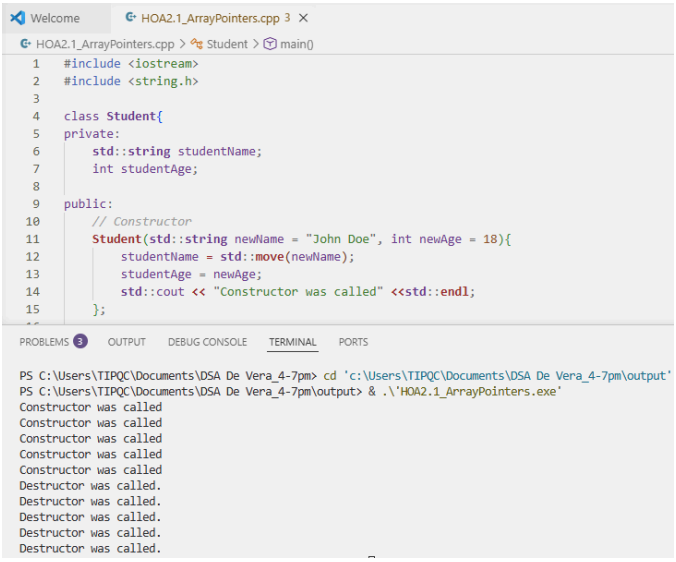
```
PS C:\Users\TIPQC\Documents\DSA De Vera_4-7pm>  & 'c:\Users\TIPQC\.vscode\extensions\ms-vscode.cpptools-1.32.2-win32-x64\debug
dfxug5.wsg' '--stdout=Microsoft-MIEngine-Out-k1riyxsr.azp' '--stderr=Microsoft-MIEngine-Error-ejaydtxe.ltq' '--pid=Microsoft-M
erpreter=mi'
Constructor was called
Copy Constructor was called.
Constructor was called
Copy Assignment Operator was called.
Destructor was called.
Destructor was called.
Destructor was called.
PS C:\Users\TIPQC\Documents\DSA De Vera_4-7pm>
```

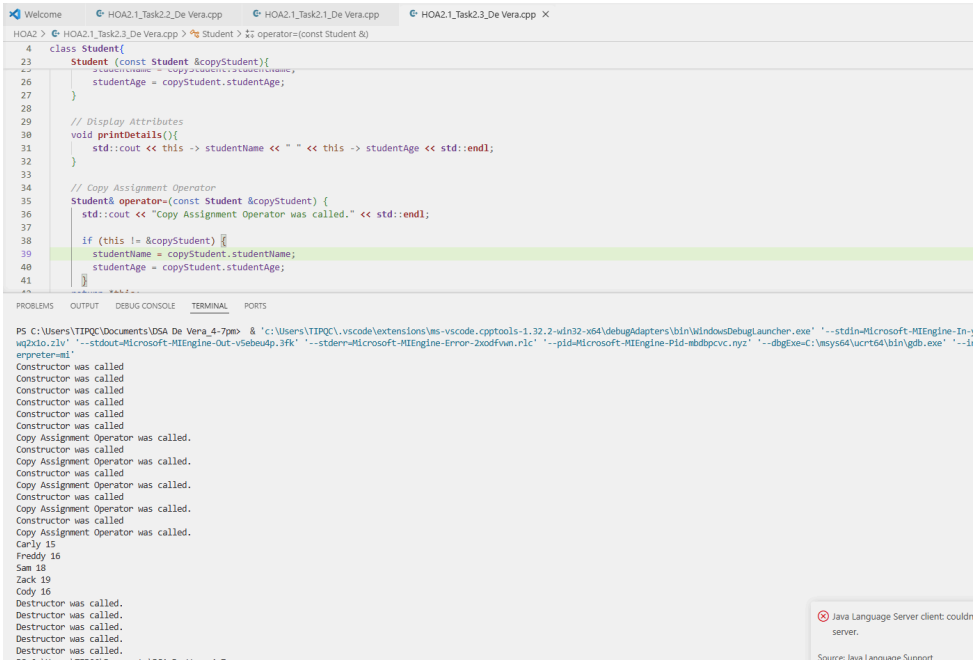
OBSERVATION

- The program demonstrates the use of a constructor, copy constructor, copy assignment operator, and destructor in a C++ class. The constructor is called when `student1` and `student3` are created, while the copy constructor is invoked when `student2` is initialized using `student1`. The copy assignment operator is used when the already existing object `student3` is assigned the values of `student2`. Finally, when the program ends, the destructor is automatically called

	for each object in reverse order of their creation, illustrating the object lifecycle in C++. -

Table 2-2. Modified Driver Program with Student Lists

SCREENSHOT	
OBSERVATION	- In this program, following the instructor's lecture and demonstration. The code must at least the guidelines of **RULES OF THREE** wherein the code defines the constructor, deconstructor, copy constructor, and/or the copy assignment operator. So as I analyze the code, it first starts with the ***constructor*** where its main job is to initialize the variables and its values. Next, the ***deconstructor***'s main task is to "destroy" the variable that is being called. Third, the ***copy constructor*** where it creates new objects using the data that is being copied in the same class. As for the last, we are instructed to make the missing ***copy assignment operator***.

SCREENSHOT	
OBSERVATION	- After I executed the program, the **CONSTRUCTOR**, **COPY CONSTRUCTOR**, **COPY ASSIGNMENT OPERATOR**, and **DESTRUCTOR** were called in a specific order based on how I wrote the program in order. Based on how I understand the code, the constructor's task in this one is to first initialize and display the **given name and age.** Then the copy constructor was called when a new object was initialized using an existing object. As for the copy assignment operator, it was executed when an existing object was assigned the values of another existing object. Lastly, the destructor was called five (5) times because five items are needed to be terminated.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

JENNY'S GROCERY LIST		
Apple	PHP 10	x7
Banana	PHP 10	x8
Brocolli	PHP 60	x12
Lettuce	PHP 50	x10

SCREENSHOT	
OBSERVATION	- Similar to the first three procedures, this supplementary activity has the **CONSTRUCTOR**, **DECONSTRUCTOR**, **COPY CONSTRUCTOR**, **DISPLAY ATTRIBUTES**, **COPY ASSIGNMENT OPERATOR**. Though the only difference and modification is the calling of the classes. The instructions also required us to create a different class for fruits and vegetables–to which I had a long time coding and understanding that part. The program also made us delete an item and deallocate it which is the "lettuce" and calculate the total of the grocery list shopping.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Google Overview and ChatGPT**

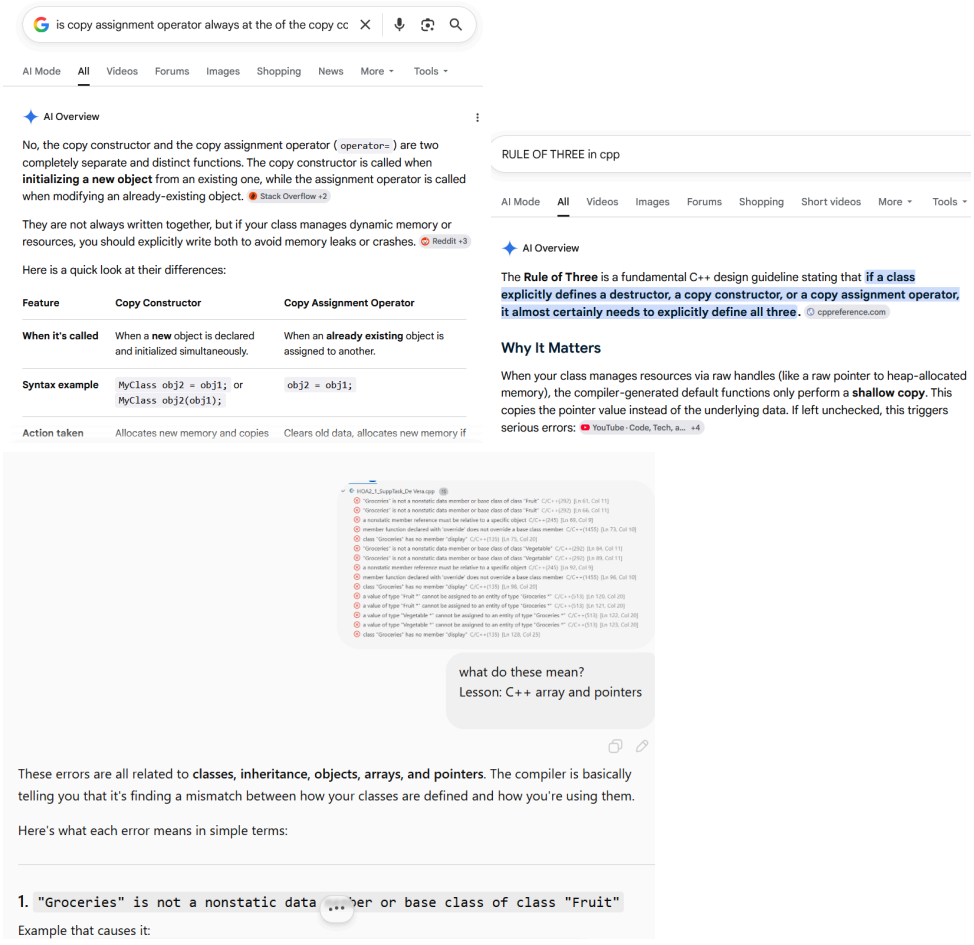
Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **To ask for clarification and clear confusion. Ask for advice and assistance.**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

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Conclusion:

After finishing this Hands-On Activity, I now understand the importance of the Arrays, Pointers and Dynamic Memory Allocation. I learned how these concepts work together to copy, manage, and destroy objects as needed. During my work process, I did have some compilation errors related to inheritance, function declarations, and class syntax. Learning and mastering the arrays and pointers can help us create and program complex applications in the future.